Explicit Formulas and Combinatorial Interpretation of Triangular Arrays via Weighted Paths

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Abstract

Using the lattice paths in $\mathbb{N} \times \mathbb{N}$, we derive a general formula for sequences $\Big(T(n,k)\Big)$ satisfying the recurrence relation of the form:

$$T((n,k) = a_{n,k}T(n-1,k) + b_{n,k}T(n-1,k-1).$$

We apply this result to the case where $a_{n,k} = a_0 + a_1k + a_2n$ and $b_{n,k} = b_0 + b_1k + b_2n$. This leads to explicit expressions for $T(n,k)$, with simpler formulas arising in the case $b_2 = 0$, as well as in the fully general case, using Faà di Bruno's type expression. In particular, we analyze the case $b_{n,k} = 1$, which frequently occurs in enumerative combinatorics. Applications include explicit formulas for the r-Eulerian numbers.We also express the case $b_{n,k} = 1$, using a transition matrix. We apply our results to several sequences.
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1 Introduction

In this paper, we investigate certain sequence of numbers that satisfy a triangular recurrence of the form

$$T(n,k) = (a_0 + a_1k + a_2n)T(n-1,k) + (b_0 + b_1k + b_2n)T(n-1,k-1). \tag{1}$$

In their book [1], Graham et al. study binomial coefficients, Stirling numbers, and Eulerian numbers, and propose a generalization problem of the form (1). We call the resulting sequences

GKP numbers. In combinatorics, several approaches have been developed to study these numbers. One of the earliest results is due to Neuwirth [5] (see also Spivey [6]), who obtained an explicit formula for the case $b_2 = 0$ using the Galton triangle. Spivey [6] further investigated several cases using finite differences. Analytical approaches have also appeared in various works, including those of Théorêt [12], Wilf [8], and Barbero [9].

Recall that a weighted set is a pair (E, v), where E is a finite nonempty set and v is a map from E to a ring of formal power series with coefficients in $\mathbb{C}$. We are generally interested in $|E|_v = \sum_{x\in E} v(x)$. Using weighted paths in the $\mathbb{N} \times \mathbb{N}$, we derive a general formula for the sequence $\Big(T(n,k)\Big)$ of the form:

$$T(n,k) = a_{n,k}T(n-1,k) + b_{n,k}T(n-1,k-1). \tag{2}$$

We apply this result to the special case of GKP numbers. Thus, the aim of this work is to study such numbers as total weights of certain paths, with initial conditions usually taken as

$$T(0,0) = 1, \quad T(n,k) = 0 \text{ if } n < \max(k,0). \tag{3}$$

We remain in the setting where $a_{n,k} = a_0 + a_1k + a_2n$ and $b_{n,k} = b_0 + b_1k + b_2n$, with $a_{n,k} \neq 0$ and $b_{n,k} \neq 0$. In particular, we study the case $b_{n,k} = 1$, which we denote mostly by $\Big(F(n,k)\Big)$. In this way, we obtain a unified approach to generalized Stirling numbers (see Hsu and Shiue [3], Maier [2]), the r-Whitney numbers (see [11]), the r-Lah numbers (see Niúl and Rácz [14]) and many others sequences in combinatorics.

However, generalizations of Eulerian numbers where $b_{n,k} \neq 1$ are also important examples when discussing sequences satisfying (1). We provide an explicit formula for $T(n,k)$ with $a_{n,k} = a_0 + a_1k + a_2n$ and $b_{n,k} = b_0 + b_1k$, and apply it on the r-Eulerian numbers.

Our Contributions and Paper Outline

We prove the main results, which provide explicit formulas for $T(n,k)$ of Faà di Bruno's type expressions:

Theorem (2.10)**.** *Let* $\Big(T(n,k)\Big)$ *be a sequence satisfying* (1) *with* $a_{n,k} = a_1k + a_0$, $b_{n,k} = b_2n + b_1k + b_0$, *and the usual initial conditions* (3)*. Then*

$$T(n,k) = \sum_{\{c_0+c_1+\cdots+c_k=n-k, 0\le c_i\le n\}} a_0^{c_0} \prod_{i=1}^{k} \Big(a_0 + a_1 i\Big)^{c_i} \Big(b_0 + (b_1+b_2)i + b_2(\sum_{j=0}^{j=i-1} c_j)\Big).$$

More generally, we also obtain the following theorem.

Theorem (2.12)**.** *Let* $\Big(T(n,k)\Big)$ *be a sequence satisfying* (1) *and the usual initial conditions* (3)*. Then*

$$T(n,k) = \sum_{\{1\le p_1<p_2<\cdots<p_{n-k}\le n\}} \prod_{i=1}^{n-k} \Big((a_2+a_1)p_i - a_1 i + a_0\Big) \prod_{i=1}^{k} \Big(b_0 + (b_1+b_2)i + b_2 \sum_{j=0}^{i-1} \#\{l : p_l - l = j\}\Big).$$

In section 2, we begin to interpret $T(n,k)$ as a total weight of a set of paths and derive explicit formula. It includes theorems 2.4, corollary 2.7 and the two above theorems. We also apply these results to sequences defined by triangular recurrence.

Section 3 contains brief remarks that we can also represent the sequences as transition matrices. We prove that the case where $b_{n,k} = 1$, it is a transition matrix in the vector space $k[x]$.

In section 4, we present a simpler formula for the case $b_2 = 0$.

2 Weighted paths in $\mathbb{N} \times \mathbb{N}$ associated with sequences of numbers satisfy a triangular recurrence

We define $\mathcal{R}_{n,k}$ as the set of paths starting from $(0,0)$ to (n,k) in the lattice $\mathbb{N} \times \mathbb{N}$, using only East steps $\mathbf{E}$ $\Big((i-1,j) \to (i,j)\Big)$ and North-East steps $\mathbf{NE}$ $\Big((i-1,j-1) \to (i,j)\Big)$. Any element $\pi \in \mathcal{R}_{n,k}$ is a path of length n and height k.

Let $a = (a_{i,j})_{i,j\in\mathbb{N}}$ and $b = (b_{i,j})_{i,j\in\mathbb{N}}$ be two sequences of real numbers. We assign a weight to each East step $\mathbf{E}$ $(i-1,j) \to (i,j)$ by $a_{i,j}$, and to each North-East step $\mathbf{NE}$ $(i-1,j-1) \to (i,j)$ by $b_{i,j}$ so one can define a valuation (weighting) $\omega_{a,b} : \mathcal{R}_{n,k} \to \mathbb{R}$, such that the weight of a path $\pi = (p_1, p_2, \cdots, p_{n+k}) \in \mathcal{R}_{n,k}$ is $\omega_{a,b}(\pi) :=$ the product of the weights of each p_i. We denote by $T_{a,b}(n,k) = |\mathcal{R}_{n,k}|_{\omega_{a,b}} = \sum_{\pi\in\mathcal{R}_{n,k}} \omega_{\mathrm{a,b}}(\pi)$ the total weight of $\mathcal{R}_{n,k}$, and by $\mathbb{T}_{a,b} = (T_{a,b}(n,k))_{n,k\in\mathbb{N}}$ the associated infinite matrix.

Lemma 2.1. *The matrix $\mathbb{T}_{a,b}$ satisfies the following recurrence relation.*

$$T_{a,b}(n,k) = a_{n,k}\mathrm{T_{a,b}}(n-1,k) + b_{n,k}\mathrm{T_{a,b}}(n-1,k-1). \tag{4}$$

It satisfies also the usual initial condition (3). *Conversely, any relation of this form, for $a_{n,k}, b_{n,k} \in \mathbb{R}$, may be interpreted as a weighting of $\mathcal{R}_{n,k}$.*

Proof. Indeed, any path from $(0,0)$ to (n,k) is either a path from $(0,0)$ to $(n-1,k)$ followed by an East step, or a path from $(0,0)$ to $(n-1,k-1)$ followed by an East step. □

Example 2.2. Figure 1 represents a path from $(0,0)$ to $(6,3)$ whose weight is $a_{1,0}b_{2,1}b_{3,2}a_{4,2}b_{5,3}a_{6,3}$.

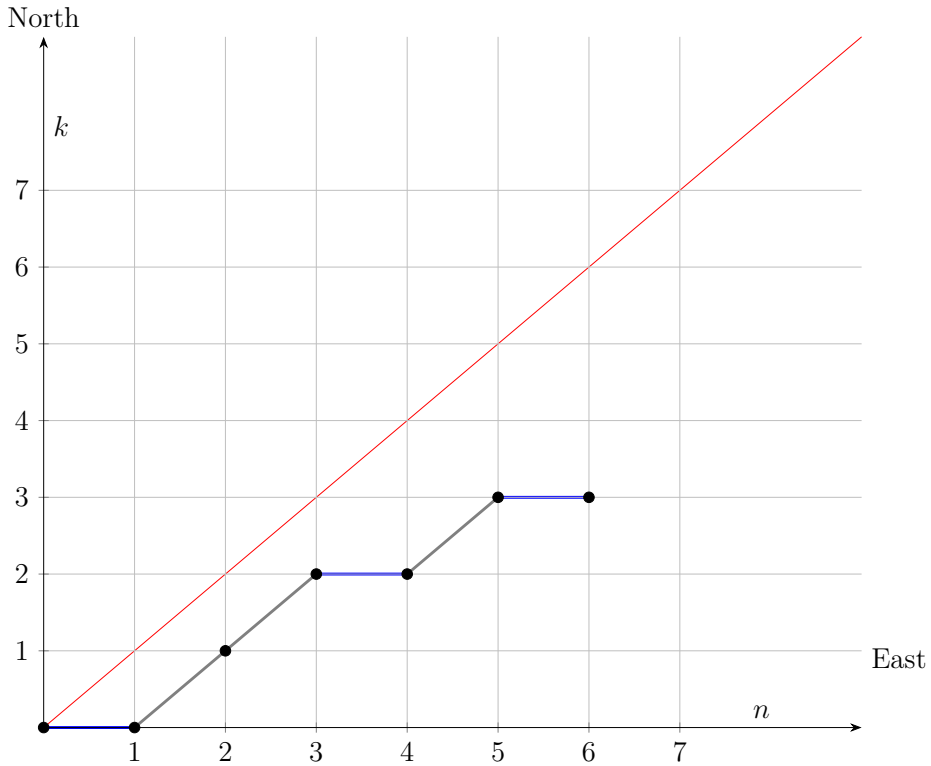

Figure 1: A path from $(0,0)$ to $(6,3)$

Let $\mathcal{C}^{\uparrow}(k,n)$ be the set of strictly increasing functions from $[k]$ to $[n]$. We identify $\pi \in \mathcal{R}_{n,k}$ with $\sigma \in \mathcal{C}^{\uparrow}(k,n)$ by defining $\sigma(i) = j$ if the i-th **NE** step is $(j-1, l-1) \to (j,l)$. Given $\sigma \in \mathcal{C}^{\uparrow}(k,n)$, one can associate the increasing list $\tilde{\sigma} = (\tilde{\sigma}_i)_{1 \le i \le n-k}$ of the elements of $[n] \setminus \sigma([k])$, and $|\mathcal{C}^{\uparrow}(k,n)| = \binom{n}{k}$.

Example 2.3. The path in Figure 1 is identified with the map σ such that $\sigma(1) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 5$, and $\tilde{\sigma} = (1,4,6)$.

Theorem 2.4. *Let $a = (a_{n,k})_{n,k \in \mathbb{N}}$, $b = (b_{n,k})_{n,k \in \mathbb{N}}$ and $T = \big(T(n,k)\big)$ be three sequences such that T satisfies the recurrence relation $T(n,k) = a_{n,k}T(n-1,k) + b_{n,k}T(n-1,k-1)$, with the initial conditions* (3)*. Then*

$$T(n,k) = \sum_{\sigma \in \mathcal{C}^{\uparrow}(k,n)} \prod_{i=1}^{n-k} \left(a_{\tilde{\sigma}_i, \tilde{\sigma}_i - i}\right) \prod_{i=1}^{k} \left(b_{\sigma(i),i}\right). \tag{5}$$

Proof. Recall our notation: $\omega_{a,b} : \mathcal{R}_{n,k} \to \mathbb{R}$, the weighting, assigns the weight $a_{i,j}$ to each East step **E** $(i-1,j) \to (i,j)$ and the weight $b_{i,j}$ to each North-East step **NE** $(i-1,j-1) \to (i,j)$. Since $T(n,k)$ is the total weight of $\mathcal{R}_{n,k}$, we have $T(n,k) = \sum_{\pi \in \mathcal{R}_{n,k}} \omega_{\mathrm{a,b}}(\pi)$.

It is straightforward to see that each East step is weighted by $a_{\tilde{\sigma}_i, \tilde{\sigma}_i - i}$ and each North-East step is weighted by $b_{\sigma(i),i}$. Thus, the result follows. $\square$

The case $b_{n,k} = 1$, corresponding to sequences $(F(n,k))$ satisfying the recurrence relation:

$$F(n,k) = F(n-1,k-1) + (a_2 n + a_1 k + a_0)F(n-1,k), \tag{6}$$

with the usual initial conditions (3), is very common in combinatorics. In this case, (5) becomes

$$F(n,k) = \sum_{\sigma \in \mathcal{C}^{\uparrow}(k,n)} \prod_{i=1}^{n-k} \Big(a_{\tilde{\sigma}_i, \tilde{\sigma}_i - i} \Big). \tag{7}$$

Since $a_{n,k} = a_2 n + a_1 k + a_0$, we obtain

Corollary 2.5. *The sequence $(F(n,k))$ satisfies the following:*

$$F(n,k) = \sum_{\{1 \le p_1 < p_2 < \cdots < p_{n-k} \le n\}} \prod_{i=1}^{n-k} \Big((a_2 + a_1) p_i - a_1 i + a_0 \Big). \tag{8}$$

Lemma 2.6. *If $b_{n,k} = b_0 + b_1 k$, then*

$$T(n,k) = (b_0 + b_1 |\, b_1\,)^{(\overline{k})} F(n,k), \tag{9}$$

where $(b_0 + b_1 |\, b_1\,)^{(\overline{k})} = (b_0 + b_1)(b_0 + 2b_1) \cdots (b_0 + k b_1)$.
In particular, if $b_1 = 0$ then:

$$T(n,k) = b_0^k F(n,k). \tag{10}$$

And if $b_0 = 0$, then:

$$T(n,k) = b_1^k k! F(n,k). \tag{11}$$

Proof. Suppose that $b_{n,k} = b_0 + b_1 k$. Since each element $\pi \in \mathcal{R}_{n,k}$ has k **NE** steps, and the weight of each of these steps is independent of n, the step at height 1 is weighted by $b_0 + b_1$, the step at height 2 by $b_0 + 2b_1$, $\cdots$, and the step at height k by $b_0 + k b_1$. Hence, the total weight of the **NE** steps is

$$(b_0 + b_1)(b_0 + 2b_1) \cdots (b_0 + k b_1).$$

This means that, for the same $a_{n,k}$, if the weight of $\pi \in \mathcal{R}_{n,k}$ is $\omega(\pi)$ in the case $b_{n,k} = 1$, and $\omega'(\pi)$ in the case $b_{n,k} = b_0 + b_1 k$, then

$$\omega'(\pi) = (b_0 + b_1)(b_0 + 2b_1) \cdots (b_0 + k b_1)\, \omega(\pi).$$

Thus we can conclude (9). □

Combining Corollary 2.5 and Lemma 2.6, we obtain

Corollary 2.7. *Let $\Big(T(n,k)\Big)$ be a sequence satisfying* (1) *with $a_{n,k} = a_2 n + a_1 k + a_0$, $b_{n,k} = b_1 k + b_0$, and the usual initial conditions* (3)*. Then*

$$T(n,k) = (b_0 + b_1 |\, b_1\,)^{(\overline{k})} \sum_{\{1 \le p_1 < p_2 < \cdots < p_{n-k} \le n\}} \prod_{i=1}^{n-k} \Big((a_2 + a_1) p_i - a_1 i + a_0 \Big). \tag{12}$$

Applications 2.8. 1. Setting $a_2 = -\alpha$, $a_1 = \beta$, $a_0 = \alpha + r$, we obtain the generalized Stirling Numbers $S_{\alpha,\beta,r}$ defined by Hsu and Shiue in [3], satisfying the recurrence relation ([3] Equation (7)):

$$S_{\alpha,\beta,r}(n,k) = S_{\alpha,\beta,r}(n-1,k-1) + (-\alpha(n-1) + \beta k + r) S_{\alpha,\beta,r}(n-1,k). \tag{13}$$

The generalized Stirling Numbers verify:

$$S_{\alpha,\beta,r}(n,k) = \sum_{\{1\le p_1<p_2<\cdots<p_{n-k}\le n\}} \prod_{i=1}^{n-k} \Big((\beta-\alpha)p_i - \beta i + \alpha + r\Big). \tag{14}$$

Note that $S_{m,0,r}(n;k)$ is the r-Whitney number of first kind, defined by Mezö in [11], and $S_{0,m,r}(n;k)$ is the r-Whitney number of second kind.

In [16], Randrianirina introduces several generalizations of the Lah numbers.

2. If we consider the sequence defined by the recurrence relation:

$$\lambda_{m,r}(n,k) = \lambda_{m,r}(n,k-1) + (m(n-1)+k+r)\lambda_{m,r}(n,k), \tag{15}$$

we have $a_0 = -m+r$; $a_1 = 1$ and $a_2 = m$. Then:

$$\lambda_{m,r}(n,k) = \sum_{\{1\le p_1<p_2<\cdots<p_{n-k}\le n\}} \prod_{i=1}^{n-k} \Big((m+1)p_i - i - m + r\Big). \tag{16}$$

$\lambda_{m,r}(n,k)$ is the matrix associated with the $\mathbb{L}$-species (see [17]) $z\mathbb{E}_\beta \circ \mathcal{Q}_m^+) \cdot \mathcal{Q}_m^r = T_{Z=z,X=x}\mathcal{G}(ZX^r)$, $\mathcal{G}$ being the combinatorial differential operator associated with the grammar $G = \{z \to \mu z x^{m+1}, x \to x^{m+1}\}$, $\mathbb{E}_\beta$ is the species of sets weithed by $w(A) = \mu^{|A|}$, and $\mathcal{Q}_m$ is the $\mathbb{L}$-species of m-Stirling permutation, weighted by $w'(\sigma) = x^{mn+1}$ if $\sigma \in \mathcal{Q}[n]$.

3. Setting $a_2 = a_1 = m$, $a_0 = r+s$, we obtain the numbers $\Lambda_{m,r,s}(n,k)$, satisfying the relation:

$$\Lambda_{m,r,s}(n,k) = \Lambda_{m,r,s}(n-1,k-1) + (m(n+k)+r+s)\Lambda_{m,r,s}(n-1,k). \tag{17}$$

We have

$$\Lambda_{m,r,s}(n,k) = \sum_{\{1\le p_1<p_2<\cdots<p_{n-k}\le n\}} \prod_{i=1}^{n-k} \Big(m(2p_i - i) + r + s\Big). \tag{18}$$

$\Lambda_{m,r,s}(n,k)$ is the matrix associated with the $\mathbb{L}$-species $\mathbb{F} = T_{V=v,U=u,X=x}\mathcal{G}(VU^sX^r)$, where $\mathcal{G}$ is the combinatorial differential operator associated with the grammar $G = \{v \to \mu v u^m x^m, u \to u x^m, x \to x^{m+1}\}$.

Note that $\Lambda_{m,r,r}(n,k)$ is the r-Whitney-Lah numbers defined by Gyimesi and Nyúl in [15], and $\Lambda_{1,r,r}(n,k)$ is the r-Lah numbers defined by Nyúl and Rácz [14])

4. If we set $a_1 = a_2 = m$ and $a_0 = mr$, we obtain the sequence $(\tau_{m,r}(n,k))$ satisfying the relation:

$$\tau_{m,r}(n,k) = \tau_{m,r}(n-1,k-1) + m(n+k+r)\tau_{m,r}(n-1,k), \tag{19}$$

and associated with $(z\mathbb{E}_\beta \circ (\mathbb{L}_\gamma)^+) \cdot (\mathbb{L}_{\gamma'})^r = T_{Z=z,X=x}\mathcal{G}(ZX^r)$, where $\mathcal{G}$ is the combinatorial differential operator associated with the grammar $G = \{v \to \mu z x^r, x \to m x^2\}$. We have

$$\tau_{m,r}(n,k) = \sum_{\{1\le p_1<p_2<\cdots<p_{n-k}\le n\}} \prod_{i=1}^{n-k} \Big(m(2p_i - i + r\Big). \tag{20}$$

To continue, let's see an useful tool. Let the sets A, B, C be defined by

$$\begin{aligned} A = \mathcal{C}^{\uparrow}(k,n) =& \{\sigma = (\sigma_1, \sigma_2, \cdots, \sigma_k) : 1 \le \sigma_1 < \sigma_2 < \cdots < \sigma_k \le n\} \quad ; \\ B = \mathcal{C}^{\uparrow}(n-k,n) =& \{\tilde{\sigma} = (\tilde{\sigma}_1, \tilde{\sigma}_2, \cdots, \tilde{\sigma}_{n-k}) : 1 \le \tilde{\sigma}_1 < \tilde{\sigma}_2 < \cdots < \tilde{\sigma}_{n-k} \le n\} \text{ and} \\ C =& \{\underline{c} = (c_0, c_1, \cdots, c_k) : \sum_{i=0}^{k} c_i = n-k, \quad \forall i, c_i \in \{0, 1, \cdots, n\}\}. \end{aligned}$$

We equip B and C with the weights β and δ defined by

$$\beta(\tilde{\sigma}) := \prod_{j=1}^{n-k} \Big(a_0 + (\tilde{\sigma}_j - j)a_1\Big) \text{ and } \delta(\underline{c}:) = \prod_{i=0}^{k} \Big(a_0 + ia_1\Big)^{c_i}.$$

Lemma 2.9. *The correspondences*

(a) $C \to A$, $\underline{c} \mapsto \sigma$ *(with* $\sigma_i = i + \sum_{j=0}^{i-1} c_j$*) and*

(b) $B \to C$, $\tilde{\sigma} \mapsto \underline{c}$ *(with* $c_i := \#\{j : \tilde{\sigma}_j - j = i\}$*)*

are well-defined bijections. Moreover (B, β) *and* (C, δ) *are isomorphic as weighted sets.*

Proof. For (a), well-definedness and injectivity are clear. Since the sets are finite and have the same cardinality[1], injectivity implies bijectivity.

For (b), the map is well defined because each c_i lies in $\{0, 1, \cdots, n\}$ and the sum $\sum c_i$ is precisely the length of $\tilde{\sigma}$, $|\tilde{\sigma}| = n - k$. Its inverse is defined recursively as follows:

- for $i = 0$: for all l with $1 \le l \le c_0$, set $\tilde{\sigma}_l = l$;
- for $i = 1, \ldots, k$: for all l with
$$\sum_{j=0}^{i-1} c_j < l \le \sum_{j=0}^{i} c_j,$$
set $\tilde{\sigma}_l = l + i$.

By construction, $\tilde{\sigma}_1 \ge 1$ and $\tilde{\sigma}_l < \tilde{\sigma}_{l+1} \le (n-k) + k = n$, so $\tilde{\sigma} \in B$. It is also clear that this is the inverse of our correspondence.

To show that it is a morphism (preserves weights), it suffices to use the definition of c_i, which counts the indices j such that $\tilde{\sigma}_j - j = i$. □

This lemma yields one of the main results of this paper. We obtain the following theorem.

Theorem 2.10. *Let* $\Big(T(n,k)\Big)$ *be a sequence satisfying* (1) *with* $a_{n,k} = a_1 k + a_0$, $b_{n,k} = b_2 n + b_1 k + b_0$, *and the usual initial conditions* (3). *Then*

$$T(n,k) = \sum_{\{c_0 + c_1 + \cdots + c_k = n-k, 0 \le c_i \le n\}} a_0^{c_0} \prod_{i=1}^{k} \Big(a_0 + a_1 i\Big)^{c_i} \Big(b_0 + (b_1 + b_2)i + b_2(\sum_{j=0}^{j=i-1} c_j)\Big). \tag{21}$$

[1]We notice it earlier $|A| = \binom{n}{k}$ and the known fact about weak composition of n in $k+1$ parts.

Proof. By Theorem 2.4, we have

$$T(n,k) = \sum_{\tilde{\sigma}\in\mathcal{C}^{\uparrow}(n-k,n)} \prod_{i=1}^{n-k} \Big(a_0 + a_1(\tilde{\sigma}_i - i)\Big) \prod_{i=1}^{k} \Big(b_0 + ib_1 + b_2\sigma_i\Big).$$

We apply Lemma 2.9 and conclude the result. □

Applications 2.11. 1. Recall that an m-Stirling permutation is a permutation $\sigma = \sigma_1\sigma_2\cdots\sigma_{mn}$ of the multiset $1^m2^m\cdots n^m$ such that for all i,j,k, if $i<j<k$ and $\sigma_i = \sigma_k$ then $\sigma_j \geq \sigma_i$. In [10], Tian-Xiao He defines the mth order Eulerian numbers of order as the number $T^{(r)}(n,k)$ of m-Stirling permutations having exactly k descents, and proves that these numbers satisfy the following recurrence relation:

$$T^{(r)}(n,k) = \Big(rn - k + (1-r)\Big)T^{(r)}(n-1,k-1) + (k+1)T^{(r)}(n-1,k), \tag{22}$$

with $B^{(r)}(n,0) = 1$ pour n≥ 1 et $B^{(r)}(n,k) = 0$ if $n \leq k$ ou $k < 0$.

We apply Theorem 2.10 to get

$$T^{(r)}(n,k) = \sum_{\{c_0+c_1+\cdots+c_k=n-k, 0\leq c_i\leq n\}} \prod_{i=1}^{k} \Big(1+i\Big)^{c_i}\Big((r-1)(i-1) + r\sum_{j=0}^{j=i-1} c_j\Big). \tag{23}$$

2. Let 's consider the r-Whitney-eulerian numbers defined by Mezö and Ramirez in [13] (see also Toufik Mansour and all in [18]) and satisfying the recurrence relation:

$$A_{m,r}(n,k) = (mk+r)A_{m,r}(n-1,k) + (mn - mk + m - r)A_{m,r}(n-1,k-1). \tag{24}$$

The r-Whitney-Eulerian numbers verify:

$$A_{m,r}(n,k) = \sum_{\{c_0+c_1+\cdots+c_k=n-k, 0\leq c_i\leq n\}} r^{c_0} \prod_{i=1}^{k} \Big(r+mi\Big)^{c_i}\Big(m - r + m(\sum_{j=0}^{j=i-1} c_j)\Big). \tag{25}$$

By a similar way, one can obtain the general case, as follows.

Theorem 2.12. *Let* $\Big(T(n,k)\Big)$ *be a sequence satisfying* (1) *and the usual initial conditions* (3). *Then*

$$T(n,k) = \sum_{\{1\leq p_1<p_2<\cdots<p_{n-k}\leq n\}} \prod_{i=1}^{n-k} \Big((a_2+a_1)p_i - a_1 i + a_0\Big) \prod_{i=1}^{k} \Big(b_0 + (b_1+b_2)i + b_2\sum_{j=0}^{i-1} \#\{l : p_l - l = j\}\Big). \tag{26}$$

Proof. First, the bijection that identifies $\tilde{\sigma} = (\tilde{\sigma}_i)_{1\leq i\leq n-k} \in \mathcal{C}^{\uparrow}(n-k,n)$ with $\sigma = (\sigma(i))_{1\leq i\leq k} \in \mathcal{C}^{\uparrow}(k,n)$ can be described as follows:

$$\sigma(i) = i + \sum_{j=0}^{i-1} \#\{l : \tilde{\sigma}_l - l = j\}.$$

Now, we can re-write Theorem 2.4 for $a_{n,k} = a_2n + a_1k + a_0$ and $b_{n,k} = b_2n + b_1k + b_0$, as

$$T(n,k) = \sum_{\tilde{\sigma}\in\mathcal{C}^{\uparrow}(n-k,n)} \prod_{i=1}^{n-k} \Big(a_2\tilde{\sigma}_i + a_1(\tilde{\sigma}_i - i) + a_0\Big) \prod_{i=1}^{k} \Big(b_2\Big(i + \sum_{j=0}^{i-1} \#\{l : \tilde{\sigma}_l - l = j\}\Big) + b_1 i + b_0\Big).$$

□

Now, let's consider the generating series $T_n(x) = \sum_{k=0}^{n} T(n,k)x^k$ and $\overline{T}_k(y) = \sum_{n\geq k} T(n,k)y^n$.

Theorem 2.13. *The generating series $T_n(y)$ and $\overline{T}_k(y)$ satisfy:*

$$T_n(x) = \Big((b_2 n + b_1 + b_0)x + a_2 n + a_0\Big)T_{n-1}(x) + (b_1 x^2 + a_1 x)T'_{n-1}(x) \tag{27}$$

and

$$(1 - (a_2 + a_1 k + a_0)y)\overline{T}_k(y) = a_2 y^2 \overline{T}'_k(y) + b_2 y^2 \overline{T}'_{k-1}(y) + (b_2 + b_1 k + b_0)y\overline{T}_{k-1}(y). \tag{28}$$

Proof. We set $\mathcal{R}_{n,*} := \bigcup_{k\geq 0} \mathcal{R}_{n,k}$, $\mathcal{R}_{*,k} := \bigcup_{n\geq k} \mathcal{R}_{n,k}$ and if $\pi \in \mathcal{R}_{n,k}$ then $l(\pi) = n$ and $h(\pi) = k$. We have

$$\begin{aligned}
T_n(x) &= \sum_{\pi\in\mathcal{R}_{n,*}} \omega_{a,b}(\pi)x^{h(\pi)} = \sum_{\pi\in\mathcal{R}_{n-1,*}} (b_2 n + b_1(h(\pi)+1) + b_0)\omega_{a,b}(\pi)x^{h(\pi)+1} \\
&\quad + \sum_{\pi\in\mathcal{R}_{n-1,*}} (a_2 n + a_1 h(\pi) + a_0)\omega_{a,b}(\pi)x^{h(\pi)} \\
&= (b_2 n + b_1 + b_0)x \sum_{\pi\in\mathcal{R}_{n-1,*}} \omega_{a,b}(\pi)x^{h(\pi)} + b_1 x^2 \sum_{\pi\in\mathcal{R}_{n-1,*}} h(\pi)\omega_{a,b}(\pi)x^{h(\pi)-1} \\
&\quad + (a_2 n + a_0) \sum_{\pi\in\mathcal{R}_{n-1,*}} \omega_{a,b}(\pi)x^{h(\pi)} + a_1 x \sum_{\pi\in\mathcal{R}_{n-1,*}} h(\pi)\omega_{a,b}(\pi)x^{h(\pi)-1} \\
&= \Big((b_2 n + b_1 + b_0)x + a_2 n + a_0\Big)T_{n-1}(x) + (b_1 x^2 + a_1 x)T'_{n-1}(x).
\end{aligned}$$

In the other hand

$$\begin{aligned}
\overline{T}_k(y) &= \sum_{n\geq k} T(n,k)y^n = \sum_{\pi\in\mathcal{R}_{*,k}} \omega_{a,b}(\pi)y^{l(\pi)} \\
&= \sum_{\pi\in\mathcal{R}_{*,k}} (a_2(l(\pi)+1) + a_1 k + a_0)\omega_{a,b}(\pi)y^{l(\pi)+1} \\
&\qquad\qquad + \sum_{\pi\in\mathcal{R}_{*,k-1}} (b_2(l(\pi)+1) + b_1 k + b_0)\omega_{a,b}(\pi)y^{l(\pi)+1} \\
&= a_2 y^2 \sum_{\pi\in\mathcal{R}_{*,k}} l(\pi)\omega_{a,b}(\pi)y^{l(\pi)-1} + (a_2 + a_1 k + a_0)y \sum_{\pi\in\mathcal{R}_{*,k}} \omega_{a,b}(\pi)y^{l(\pi)} \\
&\quad + b_2 y^2 \sum_{\pi\in\mathcal{R}_{*,k-1}} l(\pi)\omega_{a,b}(\pi)y^{l(\pi)-1} + (b_2 + b_1 k + b_0)y \sum_{\pi\in\mathcal{R}_{*,k-1}} \omega_{a,b}(\pi)y^{l(\pi)} \\
&= a_2 y^2 \overline{T}'_k(y) + (a_2 + a_1 k + a_0)y\overline{T}_k(y) + b_2 y^2 \overline{T}'_{k-1}(y) + (b_2 + b_1 k + b_0)y\overline{T}_{k-1}(y).
\end{aligned}$$

□

Noticing $T_0(x) = 1$ and if $a_2 = 0$, $\overline{T}_0(y) = \frac{1}{1-a_0 y}$, we get directly:

Corollary 2.14. *If $b_1 = a_1 = 0$, then*

$$T_n(x) = \prod_{i=1}^{n} \Big((b_2 i + b_1 + b_0)x + a_2 i + a_0\Big). \tag{29}$$

And if $a_2 = b_2 = 0$ then

$$\overline{T}_k(y) = \frac{1}{1 - a_0 y} \prod_{i=1}^{k} \frac{(b_1 k + b_0)y}{(1 - (a_1 k + a_0)y)} \tag{30}$$

These identities include the following results:

1. The r-Whitney numbers of first kind $w_{m,r}(n,k)$, corresponding to $a_2 = -m$, $a_1 = 0$, $a_0 = m - r$, $b_2 = b_1 = 0$ and $b_0 = 1$, satisfy:

$$T_n(x) = \sum_{k=0}^{n} w_{m,r}(n,k)x^k = \prod_{i=0}^{n-1}(x - r - mi) \tag{31}$$

2. The r-Whitney numbers of second kind $W_{m,r}(n,k)$, corresponding to $a_2 = 0$, $a_1 = m$, $a_0 = r$, $b_2 = b_1 = 0$ and $b_0 = 1$, satisfy:

$$\overline{T}_k(y) = \sum_{n\geq 0} W_{m,r}(n,k)y^k = \prod_{i=0}^{k} \frac{y}{(1-(mk+r)y)}. \tag{32}$$

These results are seen in several combinatorial pieces of literature ([16], [19]).

3 The sequence $(F(n,k))$ as transition matrix on the vector space $\mathbb{K}[x]$

Moreover, this approach permits us also to write some triangular arrays as transition matrix in the vector space $\mathbb{K}[x]$.

Theorem 3.1. *For any sequences $\big((F(n,k))\big)$, that satisfies the usual initial conditions*

$$F(0,0) = 1, \quad F(n,k) = 0 \text{ if } n < \max(k,0),$$

the following facts are equivalent

$$\forall n,k: \quad F(n,k) = F(n-1,k-1) + (a_2 n + a_1 k + a_0)F(n-1,k). \tag{33}$$

and

$$\forall x: \quad (x|a_2)^{(\overline{n})} = \sum_{k=0}^{n} F(n,k)(x - a_0 - a_2|a_1)^{(\underline{k})}. \tag{34}$$

where $(x|a_2)^{(\overline{n})} = \prod_{i=0}^{i=n-1}(x + ia_2)$ and $(x - a_0 - a_2|a_1)^{(\underline{k})} = \prod_{i=0}^{k-1}(x - a_0 - a_2 - ia_1)$.

Proof. Let $\mathcal{R}_n = \cup_{k\geq 0}\mathcal{R}_{n,k}$ be the set of paths of length n in $\mathbb{N}\times\mathbb{N}$, $\pi \in \mathcal{R}_n$. We define a weighting on $\mathcal{R}_n$ by setting $v(\pi) = \omega_{n,h(\pi)}(\pi)(x - a_0 - a_2|a_1)^{\underline{(h(\pi))}}$ if where $\omega_{n,k}$ is the weighting associated with $F(n,k)$ and $h(\pi)$ is the height of π. Then:

$$|\mathcal{R}_n|_v = \sum_{\pi\in\mathcal{R}_n} v(\pi) = \sum_{k\geq 0}\sum_{\pi\in\mathcal{R}_{n,k}} \omega_{n,k}(\pi)(x - a_0 - a_2|a_1)^{(\underline{k})} = \sum_{k\geq 0} F(n,k)(x - a_0 - a_2|a_1)^{(\underline{k})}.$$

Since a path π of length n can be obtained from a path u of length $n-1$ by adding either an North–East step or a East step, we have

$$
\begin{aligned}
|\mathcal{R}_n|_v &= \sum_{\pi\in\mathcal{R}_n} v(\pi) = \sum_{\pi\in\mathcal{R}_n} \omega_{n,h(\pi)}(\pi)(x-a_0-a_2|a_1)^{\underline{(h(\pi))}} \\
&= \sum_{u\in\mathcal{R}_{n-1}} (a_2 n + a_1 h(u) + a_0)\omega_{n-1,h(u)}(u)(x-a_0-a_2|a_1)^{\underline{(h(u))}} \\
&\qquad + \sum_{u\in\mathcal{R}_{n-1}} \omega_{n-1,h(u)}(u)(x-a_0-a_2|a_1)^{\underline{(h(u)+1)}} \\
&= \sum_{u\in\mathcal{R}_{n-1}} [(a_2 n + a_1 h(u) + a_0) + (x-a_0-a_2-h(u)a_1)]\omega_{n-1,h(u)}(u)(x-a_0-a_2|a_1)^{\underline{(h(u))}} \\
&= (x + a_2(n-1))|\mathcal{R}_{n-1}|_v.
\end{aligned}
$$

Noting that $|\mathcal{R}_0|_w = 1$, we obtain the result.

□

Equation (34) generalizes many results. For examples:

Applications 3.2. 1. The generalized Stirling Numbers $S_{\alpha,\beta,r}$ verify:

$$(x|-\alpha)^{(\overline{n})} = \sum_{k=0}^{n} S_{\alpha,\beta,r}(n,k)(x-r|\beta)^{(\underline{k})}. \tag{35}$$

We obtain Equation (1) of [3] (seen also in [2]). In particular, the r-Whitney numbers of first kind, verify:

$$\prod_{i=0}^{n-1}(x-mi) = \sum_{k=0}^{n} w_{m,r}(n,k)(x+r)^k. \tag{36}$$

By replacing x with mx in this equation, we obtain Equation (2) of [11]. And the r-Whitney number of second kind, verify:

$$x^n = \sum_{k=0}^{n} W_{m,r}(n,k)\prod_{i=0}^{n-k}(x-r-im). \tag{37}$$

By replacing x with $mx+r$ in this equation, we obtain Equation (1) of [11].

2. If we consider the sequence $(\lambda_{m,r}(n,k)$ defined in application 2.8: we have:

$$(x|m)^{(\overline{n})} = \sum_{k=0}^{n} \lambda_{m,r}(n,k(x-r|1)^{(\underline{k})}. \tag{38}$$

3. If we consider the sequence $(\Lambda_{m,r,s}(n,k)$, we have:

$$(x|m)^{(\overline{n})} = \sum_{k=0}^{n} \Lambda_{m,r,s}(n,k)(x-r-s-m|m)^{(\underline{k})}. \tag{39}$$

4. If we consider the sequence $\tau_{m,r}(n,k)$, with $a_1 = a_2 = m$ and $a_0 = mr$, we obtain:

$$(x|m)^{(\overline{n})} = \sum_{k=0}^{n} \tau_{m,r}(n,k)(x-mr-m|m)^{(\underline{k})}, \tag{40}$$

Remark 3.3. We note that $\Big((x|a_2)^{(\overline{n})}\Big)_{n\in\mathbb{N}}$ form a basis of $\mathbb{K}[x]$. The same holds for $\Big((x-a_0-a_2|a_1)^{(\underline{k})}\Big)_{k\in\mathbb{N}}$. Therefore, Theorem 3.1 says that $\mathbb{F}^T=\Big(F(n,k)\Big)_{k,n\in\mathbb{N}}$ is a transition matrix from$\Big((x-a_0-a_2|a_1)^{(\underline{k})}\Big)_{k\in\mathbb{N}}$ to $\Big((x|a_2)^{(\overline{n})}\Big)_{n\in\mathbb{N}}$. The matrix $\mathbb{F}^T$ is therefore invertible.

Theorem 3.4. *The inverse of $\mathbb{F}^T$, is the matrix $\mathbb{H}^T=\Big((H(n,k))\Big)$ satisfying:*

$$\forall n,k: \quad H(n,k)=H(n-1,k-1)+(-a_1n-a_2k+a_1-a_0-a_2)H(n-1,k). \tag{41}$$

Proof. Consider the inverse $\Big(H(n,k)\Big)$ of $\Big(F(n,k)\Big)$. Using (34), we have

$$\forall x: \quad (x-a_0-a_2|a_1)^{(\underline{k})}=\sum_{k=0}^{n}H(n,k)(x|a_2)^{(\overline{n})}.$$

We can change $y=x-a_0-a_2$ and use the fact $(z|c)^{(\overline{l})}=(z|-c)^{(\underline{l})}$ in this equation to get

$$\forall y: \quad (y|-a_1)^{(\overline{n})}=\sum_{k=0}^{n}H(n,k)(y+a_0+a_2|-a_2)^{(\underline{k})}.$$

Set $a_2':=-a_1;\quad a_1':=-a_2;\quad a_0':=a_1-a_0-a_2$, and plug in this equation. Then

$$\forall y: \quad (y|a_2')^{(\overline{n})}=\sum_{k=0}^{n}H(n,k)(y-a_0'-a_2'|a_1')^{(\underline{k})}.$$

So applying Theorem 3.1, we get

$$\forall n,k: \quad H(n,k)=H(n-1,k-1)+(a_2'n+a_1'k+a_0')H(n-1,k).$$

□

4 Closed formula for $F(n,k)$, and for $T(n,k)$ if $b_{n,k}=b_1k+b_0$

We now propose to prove a closed formula for the sequences $F(n,k)$. The case of the sequences $\Big(T(n,k)\Big)$, with $b_{n,k}=b_0+b_1k$ then follows as a direct consequence.

4.1 The case where $a_1\neq 0$

For this, we need the following two lemmas.

Lemma 4.1. *Let $P_n(x):=\prod_{l=1}^{l=n}(a_0+a_1x+a_2l)$. If $a_1\neq 0$, we have*

$$F(n,k)=\frac{1}{a_1^k}[x^{(\underline{k})}]P_n(x).$$

Proof. Note that $P_n(x)=(a_0+a_1x+a_2n)P_{n-1}(x)$ and $x\cdot x^{(\underline{k})}=x^{(\underline{k+1})}+k\cdot x^{(\underline{k})}$. These facts imply that $G(n,k):=[x^{(\underline{k})}]P_n(x)$ satisfies

$$G(0,0)=1;\quad G(n,k)=(a_0+a_1k+a_2n)G(n-1,k)+a_1G(n-1,k-1).$$

Using Lemma 2.6, we have $G(n,k)=a_1^kF(n,k)$. □

Lemma 4.2. *For any polynomial* $P(x) = \sum_{i=0}^{i=n} \alpha_i x^i$,

$$[x^{(\underline{k})}]P(x) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}P(j).$$

Proof. We have $x^i = \sum_{k\geq 0} S(i,k)x^{(\underline{k})}$ (see Comtet [7]). Then

$$\begin{aligned} P(x) = & \sum_{i=0}^{n}\alpha_i \sum_{k=0}^{n} S(i,k)x^{(\underline{k})} = \sum_{k=0}^{n}\Big(\sum_{i=0}^{n}\alpha_i S(i,k)\Big)x^{(\underline{k})} \\ & = \sum_{k=0}^{n}\Big(\sum_{i=0}^{n}\alpha_i \tfrac{1}{k!}\sum_{j=0}^{j=k}(-1)^{k-j}\tbinom{k}{j}j^i\Big)x^{(\underline{k})} \\ & = \sum_{k=0}^{n}\Big(\tfrac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\tbinom{k}{j}(\sum_{i=0}^{n}\alpha_i j^i)\Big)x^{(\underline{k})}. \end{aligned}$$

So the conclusion. □

Applying Lemma 4.2 to the polynomial $P_n(x)$ of Lemma 4.1, we obtain:

Theorem 4.3. *Let* $\Big(F(n,k)\Big)$ *be a sequence satisfying* (1) *with* $a_{n,k} = a_2 n + a_1 k + a_0$, $b_{n,k} = 1$, *and the usual initial conditions* (3).

If $a_1 \neq 0$, *then*

$$F(n,k) = \frac{1}{a_1^k\, k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=1}^{n}(a_0 + a_1 j + l a_2). \tag{42}$$

Applications 4.4. 1. The generalized Stirling numbers verify:

$$S_{\alpha,\beta,r}(n,k) = \frac{1}{\beta^k\, k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=0}^{n-1}(r + \beta j - l\alpha) \tag{43}$$

In particular, the r-Whitney numbers $W_{m,r}(n,k)$ of second kind satisfy:

$$W_{m,r}(n,k) = \frac{1}{m^k\, k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}(r + mj)^n. \tag{44}$$

In this case we obtain a result of Mezö and Ramirez in [13] (Corollary 8).

2. The numbers $\lambda_{m,r}(n,k)$ of Equation (15) verify:

$$\lambda_{m,r}(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=1}^{n}(r + j + ml). \tag{45}$$

3. The numbers $\Lambda_{m,r,s}(n,k)$of Equation (17) satisfy:

$$\Lambda_{m,r,s}(n,k) = \frac{1}{m^k\, k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=j}^{n+j}(r + s + ml). \tag{46}$$

In particular, the r-Whitney-Lah numbers $\Lambda_{m,r,r}$ verify:

$$\Lambda_{m,r,r}(n,k) = \frac{1}{m^k\, k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=j}^{n+j}(2r + ml). \tag{47}$$

And the r-Lah numbers $\Lambda_{1,r,r}(n,k)$ verify:

$$\Lambda_{1,r,r}(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=j}^{n+j}(2r+l). \tag{48}$$

4. The numbers $\tau_{m,r}(n,k)$ of Equation (19) verify:

$$\tau_{m,r}(n,k) = \frac{m^{n-k}}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\prod_{l=j+1}^{j+n}(r+l) = .\frac{m^{n-k}}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}\frac{(r+j+n)!}{(r+j)!} \tag{49}$$

Let's consider the generating series $F_n(x) = \sum_{k=0}^{n} F(n,k)x^k$ and $\overline{F}_k(y) = \sum_{n\geq k} F(n,k)y^k$. Then

$$F_n(x) = \sum_{\pi\in\mathcal{R}_{n,*}} \omega_{a,b}(\pi)x^{h(\pi)} = \sum_{\pi\in\mathcal{R}_{n-1,*}} \omega_{a,b}(\pi)x^{h(\pi)+1} + \sum_{\pi\in\mathcal{R}_{n-1,*}} (a_2 n + a_1 h(\pi) + a_0)\omega_{a,b}(\pi)x^{h(\pi)}.$$

Since $\sum_{\pi\in\mathcal{R}_{n-1,*}} \omega_{a,b}(\pi)h(\pi)x^{h(\pi)} = x\sum_{\pi\in\mathcal{R}_{n-1,*}} h(\pi)\omega_{a,b}(\pi)x^{h(\pi)-1}$, it follows that

$$F_n(x) = (x + a_2 n + a_0)F_{n-1}(x) + a_1 x F'_{n-1}(x). \tag{50}$$

In the other hand, setting $\mathcal{R}_{*,k} = \bigcup_{n\geq k}\mathcal{R}_{n,k}$, we have

$$\begin{aligned}
\overline{F}_k(y) &= \sum_{n\geq k} F(n,k)y^n = \sum_{\pi\in\mathcal{R}_{*,k}} \omega_{a,b}(\pi)y^{l(\pi)} \\
&= \sum_{\pi\in\mathcal{R}_{*,k}} (a_2(l(\pi)+1) + a_1 k + a_0)\omega_{a,b}(\pi)y^{l(\pi)+1} + \sum_{\pi\in\mathcal{R}_{*,k-1}} \omega_{a,b}(\pi)y^{l(\pi)+1} \\
&= a_2 y^2 \sum_{\pi\in\mathcal{R}_{*,k}} l(\pi)\omega_{a,b}(\pi)y^{l(\pi)-1} + (a_2 + a_1 k + a_0)y\sum_{\pi\in\mathcal{R}_{*,k}} \omega_{a,b}(\pi)y^{l(\pi)} \\
&\quad + y\sum_{\pi\in\mathcal{R}_{*,k-1}} \omega_{a,b}(\pi)y^{l(\pi)} \\
&= a_2 y^2 \overline{F}'_k(y) + (a_2 + a_1 k + a_0)y\overline{F}_k(y) + y\overline{F}_{k-1}(y).
\end{aligned}$$

That is

$$(1 - (a_0 + a_1 k + a_0))\overline{F}_k(y) = y\overline{F}_{k-1}(y) + a_2 y^2 \overline{F}'_k(y). \tag{51}$$

4.2 The case where $a_1 = 0$

Let us now consider the case $a_1 = 0$ and $a_2 \neq 0$. Define $Q_n(x) = \prod_{i=1}^{n}(x + a_0 + a_2 i)$. On the one hand, we have: $Q_n(x) = \sum_{k=0}^{n}\Big(\sum_{A\subset[n],|A|=n-k}\prod_{i\in A}(a_0 + i a_2)\Big)x^k$. On the other hand, we have $\prod_{i=0}^{n-1}(u+i) = \sum_{j=0}^{n} c(n,j)u^j$, where $c(n,j$ is the unsigned Stirling numbers of first kind (see Comptet [7]). Then:

$$\begin{aligned}
Q_n(x) &= \prod_{i=0}^{n-1}(x + (a_0 + a_2) + a_2 i) = a_2^n \prod_{i=0}^{n-1}(\frac{x + a_0 + a_2}{a_2} + i) = a_2^n \sum_{j=0}^{n} c(n,j)(\frac{x + a_0 + a_2}{a_2})^j \\
&= a_2^n \sum_{j=0}^{n} c(n,j)(\sum_{k=0}^{j}\binom{j}{k}a_2^{-j}x^k(a_0 + a_2)^{j-k}) = \sum_{k=0}^{n}(\sum_{j=k}^{n} c(n,j)\binom{j}{k}a_2^{n-j}(a_0 + a_2)^{j-k})x^k.
\end{aligned}$$

So

$$[x^k]Q_n(x) = \sum_{A\subset[n],|A|=n-k} \prod_{i\in A}(a_0+ia_2) = \sum_{j=k}^{n} c(n,j)\binom{j}{k} a_2^{n-j}(a_0+a_2)^{j-k}. \tag{52}$$

Let $\Big(F(n,k)\Big)$ be a sequence satisfying (1) with $a_{n,k} = a_2 n + a_0$, $b_{n,k} = 1$, and the usual initial conditions (3). The weight of each East step $\mathbf{E}$ $(i-1,j) \to (i,j)$ by $a_{i,j}$ is independent of j and is equal to $a_0 + a_2 i$ when it is the i-th step. Thus, if $\pi \in \mathcal{R}_{n,k}$, and if A_π is the set of positions of the East steps of π, then its weight is $\omega(\pi) = \prod_{i\in A_\pi}(a_0 + a_2 i)$. Since each path $\pi \in \mathcal{R}_{n,k}$ can be identified with the set A_π and $|A_\pi| = n-k$, it follows that

$$F(n,k) = \sum_{\pi\in\mathcal{R}_{n,k}} \omega(\pi) = \sum_{A\subset[n],|A|=n-j} \prod_{i\in A}(a_0+ia_2).$$

Hence we obtain

Theorem 4.5. *If $\Big(F(n,k)\Big)_{n,k\in\mathbb{N}}$ be a sequence satisfying* (1) *with $a_{n,k} = a_2 n + a_0$, $b_{n,k} = 1$, and the usual initial conditions* (3)*, then:*

$$\sum_{k=0}^{n} F(n,k)x^k = \prod_{i=1}^{n}(x + a_0 + a_2 i). \tag{53}$$

That is:

$$F(n,k) = \sum_{j=k}^{n} c(n,j)\binom{j}{k} a_2^{n-j}(a_0+a_2)^{j-k}. \tag{54}$$

Applications 4.6. 1. Let us consider the r-Whitney numbers of first kind obtained by setting $a_2 = -m$, $a_0 = m - r$. Then

$$w_{m,r}(n,k) = \sum_{j=k}^{n}(-1)^{n-k} m^{n-j} c(n,j)\binom{j}{k} r^{j-k} \tag{55}$$

We obtain a result of Randrianirina in [16] (Equation (37)).

2. More generally, let $\Big(F(n,k)\Big)$ be a sequence satisfying:

$$F(n,k) = F(n-1,k-1) + (a_0 + a_1 k)F(n-1,k). \tag{56}$$

Equation 42 say that:

$$F(n,k) = \frac{1}{a_1^k\, k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}(a_0 + a_1 j)^n. \tag{57}$$

Theorem 3.4 says that the inverse of the matrix $\mathbb{F}^T = (F(n,k))_{n,k}$, is the matrix $\mathbb{H}^T = ((H(n,k)))$ satisfying

$$H(n,k) = H(n-1,k-1) + (-a_1 n + a_1 - a_0)H(n-1,k). \tag{58}$$

We have

$$H(n,k) = \sum_{j=k}^{n} c(n,j)\binom{j}{k}(-1)^{n-k} a_1^{n-j} a_0^{j-k}. \tag{59}$$

Corollary 4.7. *Let* $\big(F(n,k)\big)$ *be a sequence satisfying* (1) *with* $a_{n,k} = a_2 n + a_1 k + a_0$, $b_{n,k} = 1$, *and the usual initial conditions* (3). *We have*

$$F(n,k) = \sum_{j=k}^{n} \left(\sum_{i=j}^{n} c(n,i) \binom{i}{j} a_2^{n-i} (a_2 + a_0)^{i-j} \right) a_1^{j-k} S(j,k). \tag{60}$$

That is

$$F(n,k) = \sum_{i=k}^{n} \sum_{j=k}^{i} \binom{i}{j} c(n,i) S(j,k) a_2^{n-i} (a_2 + a_0)^{i-j} a_1^{j-k}. \tag{61}$$

Proof. Let $\big(U(n,k)\big)$ and $\big(V(n,k)\big)$ be two triangular sequences satisfying the usual initial conditions (3), with

$$U(n,k) = (a_2 n + a_0) U(n-1,k) + U(n-1,k-1),$$

and

$$V(n,k) = a_1 k V(n-1,k) + V(n-1,k-1).$$

It is straightforward to see that $\mathbb{F} = \mathbb{U}\mathbb{V}$. On the other hand, the weights corresponding to $\big(V(n,k)\big)$ differ from those of $S(n,k)$ only by a factor of a_1^{n-k}, since $S(n,k)$ satisfies $S(n,k) = kS(n-1,k) + S(n-1,k-1)$. Hence, we have $V(n,k) = a_1^{n-k} S(n,k)$. Applying Theorem 4.5, we obtain the desired conclusion. □

4.3 The case where $a_{n,k} = a_2 n + a_1 k + a_0$ and $b_{n,k} = b_1 k + b_0$

Consider the sequence $(T(n,k))$ satisfying:

$$T(n,k) = (a_2 n + a_1 k + a_0) T(n-1,k) + (b_1 k + b_0) T(n-1,k-1) \tag{62}$$

with initial condition $T(0,0) = 1$ and $T(n,k) = 0$ if $n > k$.

We combine Lemma 2.6 with Theorems 4.3 and 4.5 to obtain the following.

Theorem 4.8. *Let* $\big(T(n,k)\big)$ *be a sequence satisfying* (1) *with* $a_{n,k} = a_2 n + a_1 k + a_0$, $b_{n,k} = b_1 k + b_0$, *and the usual initial conditions* (3).

If $a_1 \neq 0$, *then*

$$T(n,k) = \frac{(b_0 + b_1 | b_1)^{(\bar{k})}}{a_1^k \, k!} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} \prod_{r=1}^{n} (a_0 + a_1 j + r a_2), \tag{63}$$

where $(b_0 + b_1 |\, b_1\,)^{(\bar{k})} = (b_0 + b_1) \times (b_0 + 2b_1) \times \cdots \times (b_0 + k b_1)$.

If $a_1 = 0$, *then*

$$T(n,k) = (b_0 + b_1 |\, b_1\,)^{(\bar{k})} \sum_{j=k}^{n} a_2^{n-j} c(n,j) \binom{j}{k} a_0^{j-k}. \tag{64}$$

One can check that equation (64) coincides with that of Neurwirth [5]. But in general we can get the results of Neurwirth using these theorems. This is stated in Spivey [6] (equation (2)) as follows, with minor corrections and slight adjustments.

Theorem 4.9. *Let* $\big(T(n,k)\big)$ *be a sequence satisfying* (1) *with* $a_{n,k} = a_2 n + a_1 k + a_0$, $b_{n,k} = b_1 k + b_0$, *and the usual initial conditions* (3). *Then*

$$T(n,k) = (b_0 + b_1 |\, b_1\,)^{(\overline{k})} \sum_{i=k}^{n} \sum_{j=k}^{i} \binom{i}{j} c(n,i) S(j,k) a_2^{n-i} (a_2 + a_0)^{i-j} a_1^{j-k}. \tag{65}$$

This now follows directly from Lemma 2.6 and Corollary 4.7.

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